Abhinav Gupta

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Software Engineer with 6 years of experience building high-scale distributed systems, with a growing focus on AI/ML infrastructure. Designed pipelines processing 1B+ events/day, deployed ML algorithms in production, and built LLM-powered observability tooling using MCP and local LLMs.

WORK EXPERIENCE

Software Development Engineer 2 — **Bengaluru /Remote**
Media.net — **01/2025 - Present**

- Designed and implemented a unified, real-time streaming pipeline replacing multiple legacy systems, reducing data latency from ~60 min to ~5 min and improving reporting accuracy.
- Architected core modules of the company's data pipeline handling 1B+ events/day (~15K QPS) powering ML, analytics, and budgeting systems.
- Implemented and integrated a Thompson Sampling–based ad rotation system combining CTR and conversion signals, improving CTR by 10%.
- Tech Stack: Apache Flink, Aerospike, Java, Python, Argo CD, Docker, OpenTelemetry, Grafana + Prometheus

Software Development Engineer • Full-time — **Bengaluru /Remote**
Media.net — **06/2022 - 12/2024**

- Worked on a real-time contextual ad serving platform for providing high conversion rates to advertisers.
- Migrated single-ad serving to a parallel multi-ad architecture, supporting high QPS with minimal latency.
- Optimised SQL to Redis data building architecture to reduce runtime by 70%
- Created a generic system to add new ad targetings with 0 code updates.
- Worked on real-time targeting, auctioning, bidding, budgeting, logging, and analytics.
- Tech Stack: Ad-tech, Java, Node.js, Kafka, Redis, MS SQL, Cassandra, Netty

Software Development Engineer — **Bengaluru/Remote**
Reliance Jio — **08/2020 - 05/2022**

- Worked on an IOT platform for visualization and analytics of the data received from multiple types of telemetry devices.
- Implemented system for concurrent execution of microservices in different nodes.
- Containerized various microservices and servers in the project using Docker.
- Centralized logging system using an ELK stack.
- Tech Stack: Node.js, MongoDB, MongoDB Gridfs, Express, Angular.js, Apache Kafka

EDUCATION

B.E. Computer Science
Birla Institute of Technology and Science Pilani • GPA: 7.3 — Pilani, Rajasthan • 05/2020

PROJECTS

AI-Powered Budgeting Observability Assistant (MCP-based)
Independently built an internal AI assistant enabling natural language querying of real-time Cassandra data to surface live budget allocation, pacing, and delivery insights. Replaced manual CQL writing with an MCP-powered retrieval workflow — cutting debugging time from minutes of manual querying to seconds. Built using MCP and structured querying over distributed ad-tech Cassandra data.

AI Interview Preparation Assistant

Built a personalized AI interview prep assistant using locally hosted Gemma via Ollama, with context-aware adaptive problem generation for coding and system design, and persistent progress tracking.

VOLUNTEERING & LEADERSHIP

Hindi Press Club, BITS Pilani

01/2018 - 05/2018

Coordinator

- Organized a panel on Modern Journalism with eminent journalists as speakers.
- Led a team of 40+ members including writers, designers, and editors for the college technical fest, APOGEE'18.

SKILLS

Languages: Java, Python, C++, Node.js

Infrastructure & Data: Apache Flink, Kafka, Redis, Cassandra, Aerospike, MongoDB, SQL, Docker, Argo CD

Observability: Prometheus, Grafana, OpenTelemetry, ELK Stack

AI/ML Systems: Thompson Sampling, LLM Applications, MCP, Ollama

Domains: AdTech, Distributed Systems, System Design, Real-time Data Processing